A Unified Model for Intelligent Diagnosis: Synergizing Machine Learning and Reinforcement Learning

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Abstract—This study presents an intelligent AI-assisted med ical consultation system that integrates supervised machine learning and reinforcement learning for efficient and adaptive preliminary diagnosis. The system adopts a dual-module archi tecture: an Automated Diagnosis (AD) module comprising four parallel classifiers (Random Forest, XGBoost, MLP, and Transformer), and an Adaptive Symptom Dialogue (ASD) module was built on a policy gradient framework. At the same time, it innovatively combines the two modules on these basis. The LSTM+PPO (Proximal Policy Optimization) model was launched, enabling the model to be updated during inquiries and significantly improving the diagnostic accuracy rate. The core contribution of this study lies in conducting a comprehensive comparative analysis of these hybrid methods, elaborating in detail their respective architectures, feature processing techniques including Singular Value Decomposition (SVD), unique policy network training methodologies, and their final performance in diagnostic tasks. The evaluation indicators cover end-to-end accuracy rate, F1 score, average number of conversation rounds and average reward. This study combining inter-pretable classification models with dialogue-driven reinforcement learning for real-time symptom triage,showing the potential to significantly enhance diagnostic workflows, improve decision-making accuracy, and optimize the utilization of clinical resources.

***Index Terms*---Intelligent Medical Diagnosis, Reinforcement Learning, Policy Networks,Proximal Policy Optimization(PPO), Machine Learning, LSTM Networks, Transformer Networks, Sequential Symptom Elicitation.**

I. INTRODUCTION

Accurate and efficient medical diagnosis is crucial for safeguarding the well-being of patients. However, clinicians often face challenges such as information overload and time constraints in practice, and these factors may affect the accuracy and efficiency of diagnosis. Artificial intelligence (AI) and machine learning (ML) technologies offer great potential for assisting clinical decision-making. Therefore, developing strategies that can intelligently guide the collection process of this symptom in order to efficiently obtain the most diagnostically valuable information has become an urgent need. The core objective of this research is to develop an agent capable of performing interactive medical diagnoses.[3] The agent needs to sequentially inquire about symptoms and diagnose potential diseases based on the obtained information. This task faces dual challenges: not only is it necessary to achieve highly accurate disease prediction from potentially incomplete symptom information, but also to simultaneously implement an efficient consultation strategy to collect key information with the fewest interaction rounds. To address these challenges, this paper explores a hybrid approach that combines supervised learning for disease classification and reinforcement learning (RL) to optimize decision-making strategies. We developed and compared five different model architectures. These architectures respectively combine traditional classifiers such as XGBoost, random forest, and multi-layer perceptron (MLP), as well as advanced deep learning models such as Transformer and Long Short-Term Memory Network (LSTM), with policy networks. These policy networks are trained using reinforcement learning algorithms (specifically reinforce or Proximal Policy Optimization PPO) to learn the optimal consultation sequence.

II. RELATED WORKS AND EXSITING TECHNIQUES

Artificial intelligence and machine learning (ML) have been widely applied in the field of medical diagnosis, aiming to enhance the accuracy and efficiency of diagnosis.[2] Traditional machine learning methods, such as Random Forest (RF) and XGBoost, have demonstrated effectiveness in specific disease predictions based on structured data. Meanwhile, deep learning technologies, including multi-layer perceptrons (MLPS) for classification tasks, convolutional neural networks for medical image analysis, and long Short-Term memory networks (LSTMS) and Transformer models for processing serialized medical data such as electronic health records or symptom sequences, have also made significant progress. The latter is particularly good at capturing the complex dependencies among data. Despite the remarkable achievements, these techniques still face challenges such as data sparsity, model interpretability and generalization ability in clinical applications.

Medical diagnosis is essentially a sequential decision making process, the doctor through gradually collect information about the symptoms to form judgment. However, how to strike a balance between the comprehensiveness of information and the efficiency of medical consultation, and effectively handle uncertain information, remains a difficulty in current research. Reinforcement learning (RL) provides a powerful framework for solving such sequential decision-making problems, especially showing great potential in learning dialogue strategies in medical consultations. Policy gradient methods, such as the REINFORCE algorithm, and more advanced Proximal Policy Optimization (PPO) algorithms, have been applied to optimize decision sequences in medical scenarios. Previous studies have explored different state representation, action space and reward function design methods, which are crucial to the performance of RL agents.

Although the existing research, respectively discusses the different machine learning model application in the diagnosis and intensive study on strategy of learning potential, but for a variety of supervised learning models with different reinforcement learning algorithm (especially REINFORCE and PPO) combination of comprehensive comparison and analysis is still rare. This study aims to provide empirical evidence and new insights for the development of more efficient and accurate AI-assisted sequential diagnosis systems through systematic design, implementation and evaluation of five such hybrid architectures.

III. METHODOLOGY

A. Data preparing

Our dataset is a CC-BY-licensed, large-scale synthetic medical resource designed to support AI-driven diagnostic decision-making, derived from the DDXPlus Dataset [1]. It comprises comprehensive virtual patient profiles with structured clinical representations, including standardized symptom data (binary and multi categorical), ICD-10-coded disease classifications, and anonymized patient cases — all mapped to unique identifiers for interoperability. The dataset uniquely integrates differential diagnoses, non-binary symptom expressions, and symptom-disease correlations to reflect real-world clinical reasoning.This synthesis of synthetic data rigor and granular clinical knowledge provides a robust foundation for developing natural and efficient healthcare AI models.

Here are our data previews:

Data Preview 1: One row in 'release_evidences.json'

"E_91": {"name": "E_91", "code_question": "E_91", "question_fr": "Avez-vous objectiv\u00e9 ou ressenti de la fi\u00e8vre?", "question_en": "Do you have a fever (either felt or measured with a thermometer)?", "is_antecedent": false, "default_value": 0, "value_meaning": {}, "possible-values": ["V_123", "V_14", "V_15"], "data_type":"B"}, ...

× **N** samples release_evidences.json File

The dataset employs a standardized coding system to systematically record clinical consultation information. Its core mechanism involves linking a unique symptom code (e.g., E_55) to three key elements: standardized question texts with multilingual support, predefined answer option coding dictionaries (e.g., V_89 corresponding to "forehead"), and data type specifications for answers (binary/multiple-choice). Each entry forms a complete "question-answer" mapping unit, where answer options achieve cross-language consistency through dictionary-based encoding, while the "possible-values" field defines the valid response range for clinical scenarios. This system enables machine-readable conversion of symptom descriptions (e.g., pain locations), with its data structure characterized by three core features: the symptom coding system, value domain constraints, and multilingual mapping capabilities.

Data Preview 2: One row in 'release_conditions.json'

"Spontaneous pneumothorax": {"condition_name": "Spontaneous pneumothorax", "cond-name-fr": "Pneumothorax spontan\u00e9", "cond-name-eng": "Spontaneous pneumothorax", "icd10-id": "J93", "symptoms": {"E_55": {}, "E_53": {}, "E_57": {}, "E_54": {}, "E_59": {}, "E_56": {}, "E_58": {}, "E_66": {}, "E_220": {}, "E_218": {}, "E_14": {}, "E_151": {}}, "antecedents": {"E_79": {}, "E_165": {}, "E_21": {}, "E_123": {}, "E_204": {}}, "severity": 2}, ...

× **M** samples release_conditions.json File

In this document, the data adopts a structured representation format for medical knowledge, where each disease entry is stored as a JSON object containing standardized identifiers, multilingual nomenclature, and associated clinical attributes. The data structure incorporates fields for English and French disease names, ICD-10 classification codes, and employs nested dictionaries to associate symptom codes, predisposing factors, and severity assessment scores.

Data Preview 3: Random pick one row in 'train.csv'

AGE,DIFFERENTIAL_DIAGNOSIS,SEX,PATHOLOGY,EVIDENCES,INITIAL_EVIDENCE
18,"[['Bronchitis', 0.19171203430383882], ['Pneumonia', 0.17579340398940366], ['URTI', 0.16078097198012 54], ['Bronchiectasis', 0.12429044460990353], ['Tuberculosis', 0.11367177304035844], ['Influenza', 0.11057936110639896], ['HIV (initial infection)', 0.07333003867293564], ['Chagas', 0.04984197229703562]]",M,URTI,"['E_48', 'E_50', 'E_53', 'E_54_@_V_161', 'E_54_@_V_183', 'E_55_@_V_89', 'E_55_@_V_108', 'E_55_@_V_167', 'E_56_@_4', 'E_57_@_V_123', 'E_58_@_3', 'E_59_@_3', 'E_77', 'E_79', 'E_91', 'E_97', 'E_201', 'E_204_@_V_10', 'E_222']",E_9 .....

× **1025604** samples in train.csv File

The dataset ('train.csv') comprises patient demographic information (e.g., age, sex), model-predicted probabilities for multiple diseases (e.g., bronchitis, pneumonia), physician-confirmed ground truth diagnoses (e.g., upper respiratory tract infection, URTI), relevant medical queries or examinations during diagnosis (EVIDENCES), and key initial diagnostic questions (INITIAL_EVIDENCE) that guide the

model's reasoning process. This structured data facilitates comparative analysis between model predictions and actual clinical diagnoses, thereby supporting the training and optimization of medical diagnostic algorithms.

B. Data preprocessing

Our data processing begins with three core parts: a standardized disease definition library, a detailed symptom (evidence) library, and a case dataset containing the actual medical visit records of patients. The disease database and symptom database provide a unified medical concept reference for the system, respectively mapping disease names and symptom codes to unique integer ids and standardized feature names. Among them, the symptom database also distinguishes the symptoms that can be actively queried by the RL agent from the static prior information, and labels the data types of the symptoms (binary or multivalued).

For each record in the case data, we first extract demographic information such as age and gender and convert it into numerical form.

Symptom standardization and vectorization:

The list of patients' symptoms, whether in the original string form or multi-valued form (for example, the symptom "pain location" and its specific value "head"), is uniformly converted into a series of binary features. Specifically, binary symptoms use their codes directly, while multivalued symptoms expand into combinations of their "symptom codes _@_ symptom values". Subsequently, using the MultiLabelBinarizer, the complete symptom set of each patient is transformed into a high-dimensional and sparse multi-hot encoded vector. This MultiLabelBinarizer object is fitted based on the full number of symptom features to ensure the consistency of feature dimensions.

Feature Dimension Reduction (SVD):

Considering that the symptom vectors after multi-heat encoding have a relatively high dimension, in order to improve the training efficiency and generalization ability of the random forest and MLP disease prediction models, we introduce Truncated Singular Value Decomposition (Truncated SVD) for dimension reduction [14]. After the original polyfever symptom vector is combined with the numeralized demographic characteristics (age, gender), a low-dimensional dense feature representation is generated through a well-fitted SVD converter. The feature vector after this dimension reduction is the main input of the RF model.

1. Input: The preprocessed patient characteristic matrix $X \in R^{m \times n}$ (including age, gender, and symptoms encoded by polythermia).
2. Decomposition: Perform SVD decomposition on X:

$$X = U\Sigma V^T$$

3. Select K: Determine the target dimension K (for example, 300).
4. Truncation and Transformation: Use V_K (the first K right singular vectors) or $U_K\Sigma_K$ (the product of the first K left singular vectors and their corresponding singular values) to obtain the data after dimensionality reduction.

$$X_{reduced} = XV_K \in R^{m \times K}$$

After the above processing, we customized input features for different components of the system.

The input of RF and MLP disease predictors is a dense vector containing demographic and symptom information after SVD dimensionality reduction.

The known symptom set of patient i at time t

$$S_i^{(t)} = \{s_1, s_2, \dots s_k\}$$

Use the Multi-Label Binarizer converting the symptom set $S_i^{(t)}$ into a N_E-dimensional binary vector $V_{symptoms,i}^{(t)}$

$$V_{symptoms,i}^{(t)} = [v_1, v_2, \dots, v_{N_E}]$$

The complete eigenvector $X_i^{(t)}$ can be expressed as:

$$X_i^{(t)} = [Age_i, Sex_i, v_1, v_2, \dots, v_{N_E}]$$

Its dimension is $2 + N_E$.

The input of XGBoost, Transformer, LSTM disease predictor is simple standardized and vectorized data without dimensionality reduction.

All preprocessing converters and mapping relationships are fitted and constructed during the training phase and persistently saved to ensure the consistency and reproducibility of data processing during verification, testing, and actual inference.

C. XGBoost

Extreme Gradient Boost (XGBoost) was selected as a candidate model for disease probability estimators due to its outstanding performance in processing structured or tabular data, especially in classification tasks. It is an ensemble learning method based on Gradient Boosting Decision Tree (GBDT). By iteratively training new decision trees to fit the residuals (or negative gradients) of previous trees, the prediction accuracy of the overall model is gradually improved. [6]

In this system, for a given patient i, its state at a certain moment t in the conversation can be expressed as an eigenvector $X_i^{(t)}$ The XGBoost model f_{XGB} aims to learn a mapping that converts this feature vector into a probability distribution $P_i^{(t)}$ across all C candidate diseases:

$$P_i^{(t)} = [p_{i,1}^{(t)}, p_{i,2}^{(t)}, \dots, p_{i,C}^{(t)}] = f_{\mathrm{XGB}}(X_i^{(t)})$$

$p_{i,j}^{(t)}$ represent the probability that patient i has disease j at time t, and $\sum_{\mathrm{j}=1}^{\mathrm{C}} \mathrm{p}_{\mathrm{i,j}}^{(\mathrm{t})} = 1$. This probability distribution $p_{i,j}^{(t)}$ is a key input for the subsequent reinforcement learning strategy network.

D. Random Forest

Random forest, as the disease probability estimator in this system, is an ensemble learning algorithm. The core mechanism lies in constructing and combining a large number of independent decision trees to enhance the prediction accuracy and robustness of the overall model [5].

Random forests construct each decision tree by introducing randomness, which is mainly reflected in two aspects: data sampling randomness (Bagging) and feature selection randomness.

Through the two layers of randomization, every decision tree learning to different aspects of the data, so as to reduce the correlation between the trees. In the prediction stage, for classification tasks (such as disease prediction in this system), the random forest will hand over the input samples to each decision tree in the forest for judgment. The final prediction result is generated by the prediction "voting" of all trees (for example, choosing the category with the most votes, or averaging the probability of occurrence of each category).[5]

This integration strategy enables the random forest to effectively reduce the risk of overfitting and have a good tolerance for noise and outliers.

In this system, the RF model f_{RF} receive the patient feature vectors that have undergone preprocessing and SVD dimensionality reduction X_{SVD} , and output a probability distribution covering all C candidate diseases P_{RF}, provide a decision-making basis for the subsequent dialogue strategies.

E. Multi-Layer Perceptron

In the hybrid architecture of this system, we adopted the Multi-Layer Perceptron (MLP) as the core function approximator and constructed the disease probability prediction model (MLP_AD) and the dialogue policy network (PolicyNet) respectively. Multilayer perceptron is a kind of feedforward neural networks, the basic structure consists of input layer, one or more hidden layers and an output layer. Each layer contains several neurons (or nodes), and the layers are fully connected to each other through weights [7].

In each neuron, the outputs from all the neurons in the previous layer are multiplied by the corresponding weights, then summed up and a bias term is added. This result constitutes the net input of this neuron. For the JTH neuron in the l layer, its net input $z_j^{(l)}$ can be expressed as:

$$z_j^{(l)} = \sum_i (w_{ji}^{(l)} a_i^{(l-1)}) + b_j^{(l)}$$

$a_i^{(l-1)}$ is the first layer of the ith the activation of neurons in the output, $w_{ji}^{(l)}$ is to connect the two weights of neuron, $b_j^{(l)}$ is biased.

The net input of neurons is then passed through a nonlinear activation function. The implementation of the nonlinear activation function enables MLP to learn and then represent complex nonlinear relationships [7]. After forward propagation, the logits of each action are output and converted into the action selection probability through softmax function.

F. Transformer

The Transformer architecture was chosen based on its outstanding capabilities in handling complex dependencies in serialized and aggregated data, which are highly consistent with medical dialogue scenarios [8]. The self-attention mechanism enables the Transformer to dynamically weigh the correlations of all known symptoms, thereby effectively capturing long-range dependencies and complex symptom combinations [8].

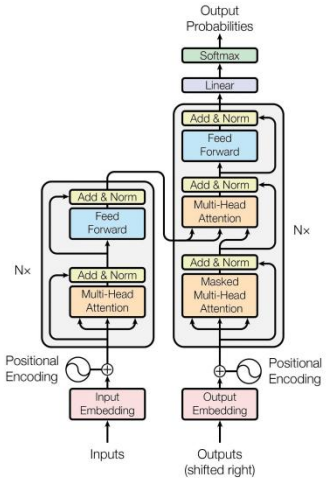


Fig.1 Transformer Model architecture

The self-attention mechanism allows the model to weigh the importance of all other elements in the sequence when encoding a specific element.

The input symptoms are first converted into dense vector representations (embeddings). Since the Transformer itself does not handle sequence order, position encoding is added to these embeddings to provide information on the relative or absolute positions of symbols in the sequence.

For each element in the input sequence, this mechanism calculates three vectors: Query Q, Key K, and Value V. The attention score between a query element and all key elements determines to what extent other value elements should be focused when representing the current query element. This is usually calculated by scaled dot-product attention:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

with d_k is the key dimension vector. Multi-Head Attention further enhances this mechanism by performing multiple attention computations with different and learned linear projections of Q, K, and V in parallel, enabling the model to simultaneously focus on information from different positions and different representation subspaces.

G. Long Short-Term Memory Network

In order to effectively capture the temporal dependencies and historical information in the medical dialogue process, this system adopts the Long Short-Term Memory (LSTM) network as the core sequence processing unit. [9]

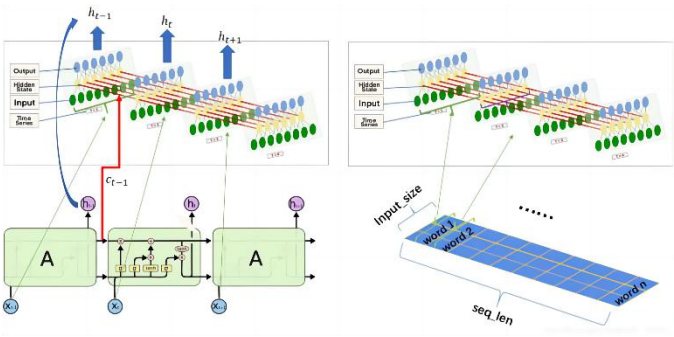


Fig.2 LSTM Model architecture

LSTM is integrated into a unified Actor-Critic model, which is responsible for policy formulation (Actor), value assessment (Critic), and auxiliary disease diagnosis (AD_Head) simultaneously. LSTM is a special kind of circulation neural network (RNN), it is by introducing ingenious gating mechanism (enter door, forgotten, output door) to solve the standard RNN in dealing with a long sequence of gradient and the gradient of disappearance of explosion problems easily.[9]

The core of LSTM is its memory unit state c, which runs through the entire sequence processing process like a conveyor belt. Information can flow on it while remaining basically unchanged, enabling the model to "remember" the information of earlier time steps.[4]

Through these gates, LSTM can selectively read, write and forget information, thereby effectively learning and utilizing long-distance context dependencies.

Medical diagnostic dialogue is essentially a sequential decision-making process. The current best question or diagnostic decision relies heavily on the interaction history of previous rounds (symptoms asked, patient responses). We choose LSTM as core sequence model, mainly based on the following consideration: in our PPO_LSTM_Model LSTM output of ht are shared with the Actor, Critic and auxiliary AD_Head. This sharing mechanism enables LSTM to learn a more comprehensive and robust dialogue state representation from multiple related task signals (policy learning, value assessment, disease diagnosis), thereby enhancing the overall performance [13].

By leveraging the powerful modeling ability of LSTM for dialogue history, we expect the system to learn more visionary and context-aware diagnostic strategies.

G.Policy Net(MLP)

In the reinforcement learning framework of this system, PolicyNet plays the role of the core decision agent. Its main responsibility is to determine the optimal action to be taken next based on the current dialogue status, in order to achieve an accurate disease diagnosis at the end of the dialogue and improve the dialogue efficiency as much as possible [2].

PolicyNet is usually implemented as a neural network. In this system, PolicyNet receives a carefully constructed State vector, State, as input. This state vector usually encapsulates the comprehensive information of the current conversation.

Based on the input State, PolicyNet outputs a probability distribution (or logits, which is later converted to probability through Softmax) $\pi(A_t|State_t;)\theta)$, where θ represents the parameters of the network.

These actions usually include:

1. Inquire about a specific and unconfirmed symptom.
2. Select "STOP" dialogue,trigger the diagnosis process.
3. Choose a disease as a diagnostic output directly.

The parameter θ of PolicyNet is trained through reinforcement learning algorithms such as REINFORCE, PPO, etc. In this way, PolicyNet learns to map states to actions that can bring long-term optimal returns, thereby achieving intelligent dialogue management and diagnostic assistance.

IV. IMPLEMENTATION

The intelligent Diagnosis System is divided into two parts: Active Symptom Discovery(ASD) and Automatic Diagnosis System(AD). The ASD part is mainly implemented by Policy Net, while the AD part exchanges different models such as XGBoost. Random Forest, MLP, Transformer, to compare the final diagnostic accuracy rate. Unlike the previous four models, in the LSTM+PPO model, we innovatively combined the ASD and AD parts into a whole, updated the parameters in a timely manner, and achieved good results.

A. XGBoost+PolicyNet

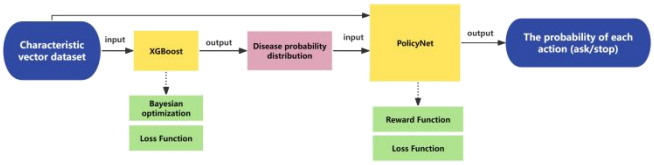


Fig.3 XGBoost+Policy Net workflow

The core interaction of the system is an iterative dialogue management process. In each round t of the dialogue, the current state $State_t$ is first constructed to input PolicyNet. This state contains two key parts of information: The first one is the multiheat encoding vector of all currently confirmed symptoms $S^{(t)}_{confirmed_vec} \in \{0,1\}^{N_E}$, N_E is the total number of symptom characteristics.The second one is based on $S^{(t)}_{confirmed_vec}$ and possibly other static characteristics such as age and gender, together constitute XGBoost input $X^{(t)}_{xgb_input}$, the probability distribution of all C candidate diseases predicted by the XGBoost model$P^{(t)}_{XGB} \in [0,1]^C$.

$$P^{(t)}_{XGB} = f_{XGB}(X^{(t)}_{xgb_input})$$

For the XGBoost model, we have made the following optimizations:

Model was trained as a multi-classification task. Let the training set be $\mathcal{D}_{train} = \{(X_i, Y_i)\}_{i=1}^{M_{train}}$, where X_i is the complete feature vector of patient i (including all known symptoms), $Y_i \in \{1, \ldots, C\}$ is the integer code of its true disease.

XGBoost builds K decision trees by optimizing an objective function that includes a loss function and a regularization term (where f_k represents the prediction of the K-th tree) :

$$\mathcal{L}(\phi) = \sum_{i=1}^{M_{train}} l\,(y_i, \hat{y}_i) + \sum_{k=1}^{K} \Omega\,(f_k)$$

$\hat{y}_i = \sum_{k=1}^{K} f_k\,(X_i)$ is the predicted output of the integrated model. For multi-classification problems, $l(y_i, \hat{y}_i)$ is usually a(multi-class logloss):

$$l(y_i, \hat{y}_i) = -\sum_{j=1}^{C} \mathbb{I}\,(Y_i = j) \log\,(p_{i,j})$$

$P_{i,j}$ are the probabilities predicted by the model that patient i belongs to category j, which are obtained by applying the Softmax function to y_i . $\Omega(f_k) = \gamma T_k + \frac{1}{2}\lambda|w_k|^2$ is the regularization term, which is used to control the complexity of the tree. T_k is the number of leaf nodes, and w_k is the weight of the leaf nodes

To obtain the best performance, this article adopts the Bayesian optimization method to systematically search the key hyperparameters of XGBoost (such as the number of trees n_estimators, the maximum depth max_depth, the learning rate learning_rate, the column sampling rate colsample_bytree, etc.). This process guides the search direction by maximizing evaluation metrics such as accuracy or AUC on the validation set [16].

We also use GPU acceleration, utilizing the library implementation such as CuPy GPU acceleration, in order to improve the training and in RL interaction efficiency of reasoning.

In the implementation, to improve efficiency, this prediction process is executed on the GPU. After appropriate tensor transformation, these two vectors are concatenated to form the input state of PolicyNet:

$$State_t = \text{concatenate}(S^{(t)}_{\text{confirmed_vec_torch}}, P^{(t)}_{\text{XGB_torch}})$$

After Policy Net receives $State_t$, output a probability distribution covering all NA feasible actions. When reasoning or interacting with users, the system usually adopts a greedy strategy to select actions $A_t = \arg\max \pi\,(a|State_t; \theta)$,And block the executed actions. If a certain symptom is selected for query, the corresponding question is raised to the environment (or user), and $S_{\text{confirmed}_{\text{vec}}}$ is updated based on the feedback. This process repeats itself until PolicyNet selects the "stop" action or reaches the preset maximum number of dialogue rounds.

PolicyNet training follow the reinforcement learning paradigm, adopted REINFORCE strategy gradient methods. In each training round, PolicyNet interacts with a simulated environment. After each action At is executed, the environment will reward according to the preset reward function $R(State_t, A_t)$ Return a scalar reward R_t. This reward function comprehensively considers factors such as the validity of symptom queries, the accuracy of the final diagnosis, the number of rounds of conversations, and the timing of stopping.

After one round ends, the strategy gradient loss is calculated using the trajectories collected during this round. The goal of the REINFORCE algorithm is to maximize the expected cumulative reward [10].

$$J(\theta) = \mathbb{E}_{\tau\sim\pi_\theta(\tau)}[\sum_{t=0}^{T} \gamma^t R_t]$$

γ is the discount factor. The loss function is usually defined as

$$\mathcal{L}(\theta) = -\sum_{\tau}\left(\sum_{t=0}^{T} \nabla_\theta \log \pi\,(A_t|State_t; \theta) \cdot G_t\right),$$

$$G_t = \sum_{k=t}^{T} \gamma^{k-t} R_k$$

The discount accumulation rewards start from the time step t . Update the parameter θ of PolicyNet through an optimizer (such as AdamW, in conjunction with a learning rate scheduler). This process is iterated through multiple cycles (epochs) until the performance of PolicyNet converges.

At the end of the dialogue process, the system will, based on the final confirmed symptom set, call the XGBoost model again to obtain the final disease probability distribution $P^{(final)}_{XGB}$,diseases with the highest probability are regarded as the diagnostic output of the system.In this way, the probability prediction ability of XGBoost is closely combined with the sequential decision-making ability of PolicyNet, jointly driving the dialogue to advance towards the goal of efficient and accurate diagnosis.

```
Algorithm: XGBoost_PolicyNet_DenseReward

FUNCTION fit_xgboost():
    best ← bayes_optimize(xgb_cv)
    xgb ← XGBClassifier(**best).fit(X_300, y)
END

FUNCTION xgb_prob(x):
    p ← xgb.predict_proba(x) + ε
    RETURN softmax(p / T)
END

FUNCTION dense_reward_RL():
    FOR epoch = 1 .. EPOCHS DO
        FOR batch IN data_chunks DO
            bonus ← symptom_hits * BONUS_EVID
            IG ← IG_WEIGHT * (H0 - H1)
            R ← IG - step_penalty + final_reward + bonus
            loss ← -logπ * norm(R) - ENT_A * entropy
            backprop(loss)
        END
        scheduler.step()
    END
END
```

B. Random Forest + Policy Net

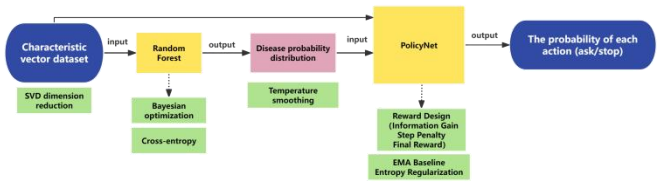


Fig.4 Random Forest + Policy Net workflow

Firstly, the dimensionality reduced vector $X_{SVD}^{(t)}$ is sent into the pre-trained random forest model to predict the probability distribution $P_{RF}^{(t)}$ of C possible diseases. The original probability distribution is further processed by temperature smoothing (temperature parameter T) to obtain the adjusted probability $P_{RF}^{(t)}$ ：

$$P'_{RF,j}(t) = \frac{\exp\left((P_{RF,j}^{(t)})/T\right)}{\sum_{k=1}^{C} \exp\left((P_{RF,k}^{(t)})/T\right)}$$

Ultimately, the input state of PolicyNet is derived from the dimensionally reduced feature vector $X_{SVD}^{(t)}$ and the smoothest RF disease probability $P_{RF}^{(t)}$ concatenated and may also contain auxiliary coding information such as the number of dialogue rounds.

PolicyNet receives the State and outputs a probability distribution covering all feasible actions (including querying symptoms and "stopping" conversations). When reasoning, the system adopts a greedy strategy to select actions and avoids repetitive questioning.

The training of PolicyNet adopts the REINFORCE policy gradient method. The agent interacts with the simulated environment in each training round, and the environment returns A carefully designed scalar reward Rt based on the action A it performs in the state. The objective of this reward function R is to guide the agent to learn a dialogue strategy that can not only efficiently collect information but also ultimately achieve accurate diagnosis. Its composition consists of multiple core components to balance the learning goals in different aspects:

First, in order to drive the agent to learn towards the final diagnostic accuracy, when the conversation ends, the system will evaluate the correctness of the final diagnosis. If the prediction is correct, avoid punishment. If the prediction is incorrect, a significant negative reward will be imposed. This is the basis for guiding agents to learn the correct diagnostic logic. Secondly, in order to promote efficiency of dialogue, in every step is introduced into a small negative rewards, to encourage agent less rounds as possible to complete the task. Meanwhile, to ensure the effectiveness of the conversation, if the agent chooses to repeatedly ask about the confirmed symptoms, it will be subject to additional penalties for repeated questioning.

$$R_{efficiency} = -\text{STEP_PENALTY} - \mathbb{I}(\text{is_duplicate}) \cdot \text{DUP_PENALTY})$$

Where $\mathbb{I}(\text{is_duplicate})$ is an indicator function. It is 1 when the question repeats; otherwise, it is 0.

Most crucially, to encourage agents to actively explore and raise the most diagnosticically valuable questions, we have introduced Information Gain (IG) rewards. This reward program aims to quantify the contribution of a problem to reducing the current diagnostic uncertainty. The calculation of information gain may be based on the entropy change of the disease probability distribution output by the RF model or other relevant metrics before and after inquiring about a certain symptom. The weight w_{IG} of this information gain reward is not fixed but dynamically adjusted with the training period e, linearly increasing from the initial value to the final value over a period of RL_EPOCHS:

$$w_{IG}^{(e)} = IG + (IG - IG) \times \frac{e}{RL}$$

This dynamic adjustment strategy enables the agent to focus more on mastering the basic dialogue and diagnostic processes in the early stage of training, while in the later stage, it pays more attention to enhancing the "intelligence" of questioning, that is, choosing the questions with the largest amount of information. Therefore, when the agent selects a value with a specific information gain IG ，The corresponding reward it will obtain is

$$R_{IG} = w_{IG}^{(e)} \times IG$$

For each time step t, the instantaneous total reward Rt obtained by the agent is the synthesis of the above components When the policy gradient is updated, the discount accumulation reward starting from the time step t is used

$$G_t = \sum_{k=t}^{T} \gamma^{k-t} R_k$$

γ is the standard time discount factor.

Through this series of elaborately designed reward signals, PolicyNet is guided to learn a dialogue strategy that can achieve an excellent balance among accuracy, dialogue efficiency and the quality of information acquisition.

```
Algorithm: RandomForest_PolicyNet

FUNCTION preprocess():
    load csv, one-hot encode
    svd ← TruncatedSVD(300).fit(X)
    X_300 ← svd.transform(X)
END

FUNCTION train_random_forest():
    best_hp ← bayes_optimize(rf_cv)
    rf ← RandomForest(**best_hp).fit(X_300, y)
END

FUNCTION rf_policy_dist(x, T):
    p ← rf.predict_proba(x) + ε
    p ← p ** (1 / T)
    RETURN p / sum(p)
END

FUNCTION train_policy_net():
    FOR epoch = 1 .. RL_EPOCHS DO
        FOR batch IN data_chunks DO
            R ← compute_dense_reward(batch)
            loss ← -logπ * (R - baseline) - α * entropy
            backprop(loss)
            scheduler.step()
        END
    END
END
```

C. *Multi-Layer Perceptron+ Policy Net*

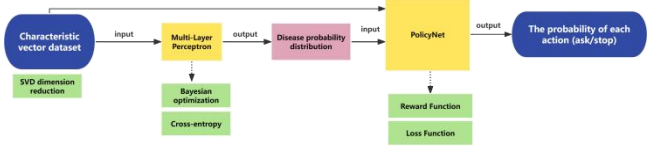


Fig.6 Multi-Layer Perceptron + Policy Net workflow

This model include a disease prediction model (MLP_AD) based on multi-layer perceptron (MLP), a reinforcement learning policy network (PolicyNet) also based on MLP, and a standard library for data processing and model management.

The characteristics of the patient (D_{demo} dimension) are combined with the multihot encoded vector of the current confirmed symptoms (N_E dimension) to form

$$D_{full} = D_{demo} + N_E$$

.

After dimension reduction by SVD, the output is the characteristics of the tensor $X_{SVD}^{(t)}$. This dimension reduction of tensor $X_{SVD}^{(t)}$ was taken to advance training PyTorch MLP disease prediction model, the internal output C possible disease logits model.

These logits are then transformed into the probability distribution through the Softmax function, combined with temperature smoothing T. Ultimately, the input state Statet of PolicyNet is the input of dimension

$$D_{state} = K_{SVD} + C + D_{round_enc}$$

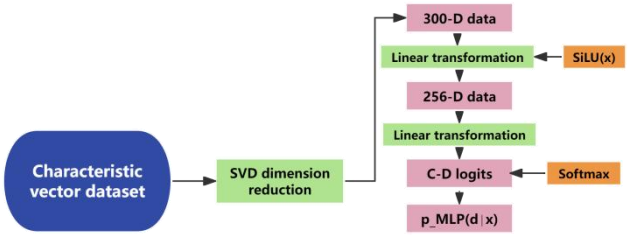


Fig.5 Data dimension transformation in MLP

PolicyNet itself is also an MLP (Dimensional Action logits), outproducing the probability of action.

In order to optimize the training process, we adopted the following key technologies.

Vector dynamic scheduling: Use warm-up + cosine Warm Restarts strategy, at the beginning of the training will be more small linear preheat to peak value, then gradually after annealing, conducive to the stability model of exploration and fast convergence. [16]

Normalization of advantages: For each batch discount accumulated award within Gt standardizing

$$G_t^{textnorm} = (G_t - mu_G)/(sigma_G + epsilon)$$

to reduce the gradient estimation variance.

Dynamic entropy regularization: To encourage exploration, add strategy in loss function item entropy $H(\pi(\cdot|State_t;\theta))$ The coefficient $\beta_t^{(e)}$ linearly decays from the initial value with the training period e to balance exploration and utilization.

$$\mathcal{L}(\theta) = -\mathbb{E}_{\tau\sim\pi_\theta}\left[\sum_{t=0}^{T}\left(\log\pi(A_t|State_t;\theta)\cdot G_t^{\text{norm}} + \beta_t^{(e)}H(\pi(\cdot|State_t;\theta))\right)\right]$$

Hybrid precision training: By using GradScaler and autocast, some operations are allowed to use half-precision floating-point numbers to accelerate training and reduce memory usage.

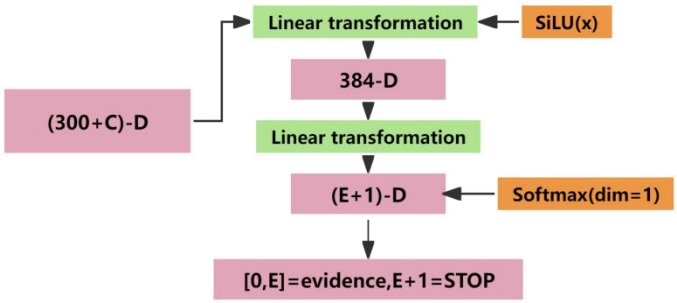


Fig.7 Data dimension transformation in Policy Net

We through TorchSVD custom class to seamlessly integrated into PyTorch SVD function tensor operation process, implements the end-to-end PyTorch model architecture; And in the PolicyNet training, advanced technologies such as learning rate scheduling, advantage normalization, dynamic entropy regularization and mixed precision training were comprehensively applied. Combined with the refined information exploration reward mechanism, they jointly ensured the efficient training and excellent performance of the system.[19]

```
Algorithm: MLP_AD_PolicyNet_Offline

FUNCTION train_AD_MLP():
    FOR ep = 1 .. AD_EPOCHS DO
        loss ← CE(AD_MLP(SVD(x)), label)
        backprop(loss)
    END
END

FUNCTION offline_REINFORCE():
    FOR ep = 1 .. RL_EPOCHS DO
        FOR batch IN data_chunks DO
            logπ_seq, ent_seq ← []
            bonus ← evidence_bonus(batch)
            R ← success_reward + bonus − penalty
            adv ← (R − mean) / std
            loss ← −sum(logπ_seq) * adv − α * sum(ent_seq)
            backprop(loss)
        END
        scheduler.step()
    END
END
```

D. Transformer+ Policy Net + PPO Algorithm

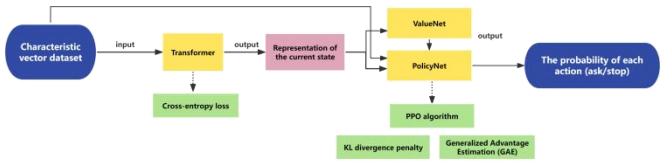


Fig.8 Transformer + PPO model workflow

This paper implemented the medical dialogue system based on the Transformer architecture and Proximal Policy Optimization (PPO) [11]. It takes advantage of the powerful capture ability of the Transformer model for serialized information and context dependence in medical conversations, and combines the PPO algorithm for efficient and stable policy learning. All components, including the disease prediction model, the Dialogue policy network, and its associated value network (V-Net), all adopt the Transformer encoder as their backbone structure.

In each round t of the dialogue, the features input to AD-Net, Pi-Net and V-Net are all based on the same context information, but serve different goals. After these features are combined with position encoding, they are sent to their respective Transformer encoder layers.

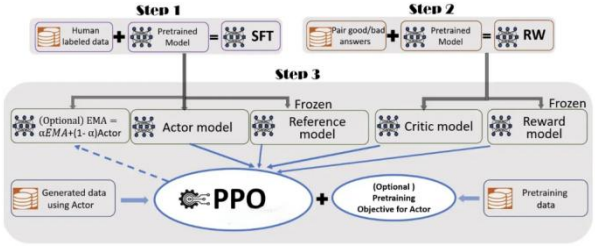


Fig. 10 PPO Algorithm Architecture

The training of Pi-Net and V-Net adopts the Proximal Policy Optimization (PPO) algorithm, which is an advanced Actor-Critic method and is renowned for its sampling efficiency and training stability in complex environments. The key implementation technologies include:

Generalized Dominance Estimation (GAE): To accurately evaluate the relative quality of each action, we use GAE to calculate the dominance function[15]

$$\widehat{A}_t = \sum_{l=0}^{T-t-1} (\gamma\lambda)^l \delta_{t+l}$$

where T is the TD error, γ is the discount factor, and λ is the smoothing parameter of GAE.

PPO Clipped Surrogate Objective and multiple rounds of updates: The loss function of Actor (Pi-Net) adopts the signature clipped surrogate objective of PPO, by restricting the probability ratio $r_t(\theta)$ of the old and new policies is within a cropping range $[1-\epsilon, 1+\epsilon]$ to prevent excessive steps in policy updates and ensure the stability of training.

$$L^{CLIP}(\theta) = \widehat{\mathbb{E}}_t\left[\min\left(r_t(\theta)\widehat{A}_t, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\widehat{A}_t\right)\right]$$

Meanwhile, by taking advantage of the feature of PPO that allows multiple gradient updates based on the experience collected in the same batch, the data utilization efficiency is significantly improved. Critic (V-Net) is updated by minimizing the mean square value of the TD error and supplemented by entropy regularization to encourage policy exploration [12].

Furthermore, we also adopt the learning rate WARM_UP and linear attenuation strategies. To precisely control the extent of strategy update, we have introduced a dynamic penalty mechanism based on KL divergence. By monitoring the KL divergence between the old and new strategies and comparing it with a target KL value, the KL penalty coefficient β KL is dynamically adjusted. This item is added to the loss function or used to stop the internal update loop in advance to ensure that the strategy does not deviate too far.

This paper still use the reward function to balance diagnostic accuracy, conversation efficiency and information acquisition. It not only includes success/failure rewards based on the final AD-Net prediction accuracy and regular step penalties, but also particularly emphasizes information gain (IG) rewards, whose weights can be dynamically adjusted to guide the agent to learn to raise the most discriminatory questions. At the end of the conversation, AD-Net gives the diagnostic probability based on the finally confirmed symptoms.

This implementation based on Transformer and PPO, its core innovation lies in: the deep modeling of complex dependencies in medical conversations by the all-Transformer architecture; The PPO algorithm and its KL divergence control variant ensure stable and efficient exploration and learning in a highly complex policy space. And through engineering optimization methods such as parallelization and mixed-precision training, efficient processing of large-scale dialogue experiences has been achieved. The combination of these elements enables the system to learn more intelligent and context-aware diagnostic dialogue strategies.

```
Algorithm: PPO_v6_4_TransformerAD

FUNCTION build_encoder():
    linear_in(2 + E) → Transformer(N layers) → cls_head(C)
END

FUNCTION init_actor_critic():
    actor ← MLP(2+E+C, E+1)
    critic ← MLP(2+E+C, 1)
END

FUNCTION pretrain_AD():
    FOR ep = 1 .. AD_EPOCHS DO
        loss ← CE(encoder(x), label)
        backprop(loss)
    END
END

FUNCTION ppo_dynamic_KL():
    FOR epoch = 1 .. ASD_EPOCHS DO
        traj ← collect_rollouts()
        adv, ret ← GAE(traj)
        FOR k = 1 .. K_EPOCHS DO
            KL ← compute_KL(old_logπ, new_logπ)
            loss ← clip_loss + 0.5*value_loss + KL_penalty(KL) - ent_coef*entropy
            backprop(loss)
        END
        update_beta_alpha(KL)
        lr_schedule.step()
    END
END
```

E. Long Short-Term Memory Network + PPO Algorithm

Different from the previous four models, this model integrates the inquiry part and the analysis part together. After each step of inquiry, it continuously optimizes its own parameters, significantly improving the accuracy of the

inquiry. The core of this system is a unified PO-LSTM model, which integrates Actor (Policy network), Critic (Value network), and an auxiliary disease prediction head (AD_Head). This integrated architecture allows the model to process the dialogue history in sequence, which is significantly different from those models that treat each round of state as independent or use encoders that are less sensitive to history.

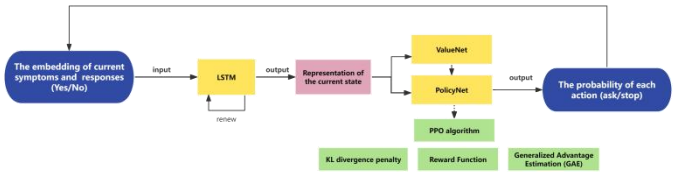


Fig.11 LSTM + PPO model workflow

At each step t of the dialogue, the model receives an index of the evidence from the previous question and a binary flag indicating the patient's response. This information, along with the patient's static characteristics (age and gender) processed through the embedding layer, is first preliminarily integrated through an input MLP. This integrated feature is then sent to the core LSTM layer. LSTM layer receive input and a moment to hide status (h_{t-1}, c_{t-1}), and the output of the current moment hidden state (h_t, c_t). The hidden state of this LSTM constitutes a rich and memorable representation of the conversation history to date.

$$(h_t, c_t) = \mathrm{LSTM}(\mathrm{MLP}_{\mathrm{input}}(\mathrm{idx}_{\mathrm{ev}_t}, \mathrm{ans}_t, \mathrm{sex}, \mathrm{age}), (h_{t-1}, c_{t-1}))$$

The hidden state h_t is then sent to three decoupled output headers: an Actor header outputs that cover the logits of all N_A possible actions (including symptom query, "stop inquiry" after E pieces of evidence, and diagnosis of C diseases); A Critic header outputs the value estimate V $V(State_t)$of the current state; And an auxiliary disease prediction head outputs the logits of C diseases. AD_Head conducts concurrent training by using binary cross-entropy (BCE) loss on real disease labels, which helps LSTM learn high-quality representations that are more relevant to diagnosis.

Similar to the previous model, the training of the policy (Actor) and value (Critic) functions adopts the Proximal Policy Optimization (PPO) algorithm. PPO was chosen because of its ability to balance sampling efficiency and learning stability by limiting the update amplitude of the strategy. In each training cycle, the system collect interactive trajectory, and use the generalized advantage (GAE) Actor the loss function of PPO landmark cut in alternative goals (Clipped Surrogate Objective) :

$$L^{\mathrm{CLIP}}(\theta) = \hat{\mathbb{E}}_t\left[\min\left(r_t(\theta)\hat{A}_t, \mathrm{clip}(r_t(\theta), 1-\epsilon_{\mathrm{clip}}, 1+\epsilon_{\mathrm{clip}})\hat{A}_t\right)\right]$$

The loss of Critic is the mean square error between its value prediction and empirical return.

To encourage exploration, the reward item of the policy entropy (is added to the loss of the Actor). Like a model, the implementation of the same process of PPO training for fine control. We adopted the dynamic KL divergence penalty mechanism: By monitoring the KL divergence between the old and new strategies and comparing it with a target KL value, the KL penalty coefficient was dynamically adjusted to prevent excessive policy updates from deviating from the stable region. Meanwhile, the linear decay strategy of the learning rate and the mixed precision training were implemented (to accelerate the training on CUDA devices and reduce the consumption of video memory) [17].

Reward function R is designed to guide the agent to study effective dialogue. It includes rewards for successfully inquiring about symptoms related to the real disease, fixed penalties for each question, and significant rewards or penalties based on the final prediction accuracy of AD_{HEAD}.

The core innovation of this implementation based on LSTM and PPO lies in: the unified model architecture centered on LSTM can effectively capture and utilize the sequence representations throughout the entire conversation, and improve the representation quality in combination with auxiliary diagnostic tasks; The combination of the PPO algorithm and advanced training techniques such as dynamic KL divergence control ensures stable and efficient policy learning in complex states and action Spaces .

```
Algorithm: PPO_LSTM_AMP_BC

FUNCTION build_vocab():
    load csv → rows
    build evidence maps (e2i, i2e)
    build condition maps (c2i, i2c)
END

FUNCTION init_model(E, C):
    create ev_emb(E, emb), ans_emb(2, emb)
    lstm_cell ← LSTMCell(emb, hid)
    actor ← Linear(hid, E + C)
    critic ← Linear(hid, 1)
END

FUNCTION behavior_cloning(model, loader, epochs):
    FOR epoch = 1 .. epochs DO
        FOR batch IN loader DO
            h,c ← model.init_state(sex, age)
            FOR ev IN batch.seq DO
                h,c,_ ← model.step(ev.idx, 1, h, c)
            END
            logits ← model.actor(h)[E:]
            loss ← CE(logits, label)
            backprop(loss)
        END
    END
END

FUNCTION ppo_train(model, rows, iterations):
    FOR it = 1 .. iterations DO
        trajs ← rollout_batch(model)
        adv, ret ← GAE(trajs, γ, λ)
        FOR k = 1 .. ppo_epochs DO
            loss ← PPO_CLIP(trajs, adv, ret, ε)
            backprop(loss)
        END
        adjust_beta_KL()
    END
END
```

V. EXPERIMENTAL AND PERFORMANCE ANLAYSIS

This chapter aims to conduct a comprehensive assessment and in-depth analysis of the performance of the five different modeling methods we have explored for the medical dialogue system. The key performance indicators include diagnostic accuracy rate, precision rate, recall rate, F1 score, area under the ROC curve (ROC AUC), average

number of dialogue rounds, as well as the average reward of the reinforcement learning agent and resource consumption during the training process.

A. XGBoost+PolicyNet

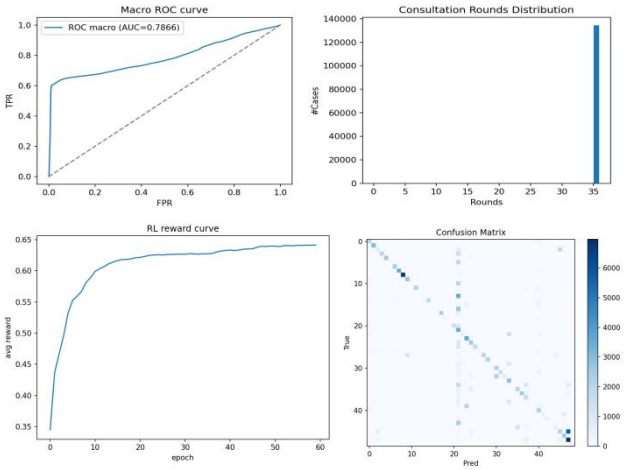


Fig. 12: XGBoost performance result

The accuracy rate of this model is 0.5866, the F1 score is 0.5435, and the ROC AUC is 0.7893. The AUC value shown by the ROC curve is 0.7893, indicating that it has a certain discriminatory ability, but there is still room for improvement. Most of the conversations lasted until nearly the maximum number of rounds, indicating that the strategy was not efficient in quickly collecting key information.

XGBoost as a powerful classifier, provides a solid foundation for disease prediction. However, the high variance of the REINFORCE algorithm may limit the efficiency and optimality of policy network learning, resulting in an excessive number of dialogue rounds.

B. Random Forest + Policy Net

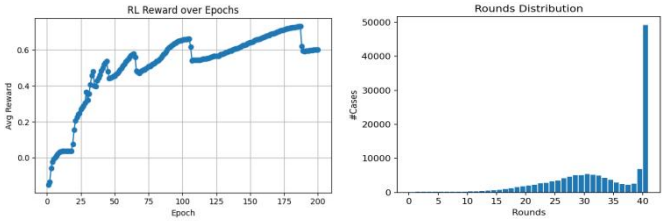


Fig. 13: Random Forest performance result

The accuracy rate was 0.4152, the F1 score was 0.343, and the ROC AUC was 0.7167. The model has a weak discrimination ability in multiple categories and a high degree of confusion. Most of the inquiry rounds have reached the maximum number. Introduction of SVD dimension reduction aims to optimize the processing of high-dimensional sparse characteristics, dynamic IG weighted reward is to promote strategy intelligent attempt.

Judging from the results, the basic diagnostic capability of the RF model under this setting may be inferior to that of XGBoost, or the SVD dimension reduction has lost key information, resulting in poor overall performance. The limitations of the REINFORCE algorithm may also exacerbate this problem.

C. MLP + Policy Net

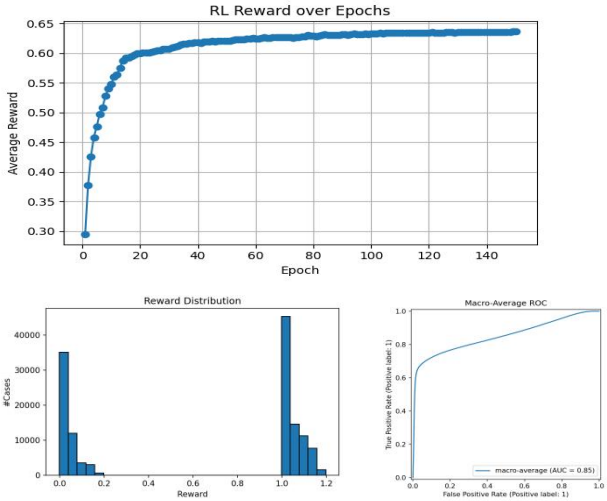


Fig. 14: MLP performance result

The accuracy rate was 0.5975, the F1 score was 0.5543, and the ROC AUC was 0.8478. These indicators are similar to the XGBoost model and even slightly better in AUC.

Although the diagnostic performance is acceptable, the dialogue efficiency is still limited by the REINFORCE algorithm. The MLP model may require more refined regularization compared to the tree model to avoid overfitting.

D. Transformer + Policy Net + PPO

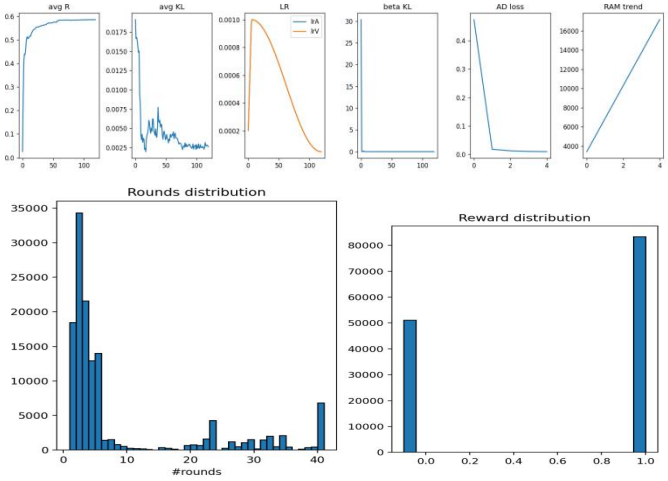


Fig. 15: Transformer performance result

The accuracy rate is 0.6196 and the F1 score is 0.6165. The Transformer model demonstrates better diagnostic performance than the previous three models. The average number of dialogue rounds decreased significantly to 8.64 rounds. The distribution is highly concentrated in a small number of rounds, indicating that the strategy is efficient.

The powerful context modeling ability of the Transformer and the stability and sample efficiency of the PPO algorithm are the keys to the performance improvement

of this model. Dynamic KL divergence control further ensures the stability of training. The significantly reduced number of dialogue rounds proves the superiority of the strategy, but high resource consumption is its main bottleneck.

E. LSTM + PPO

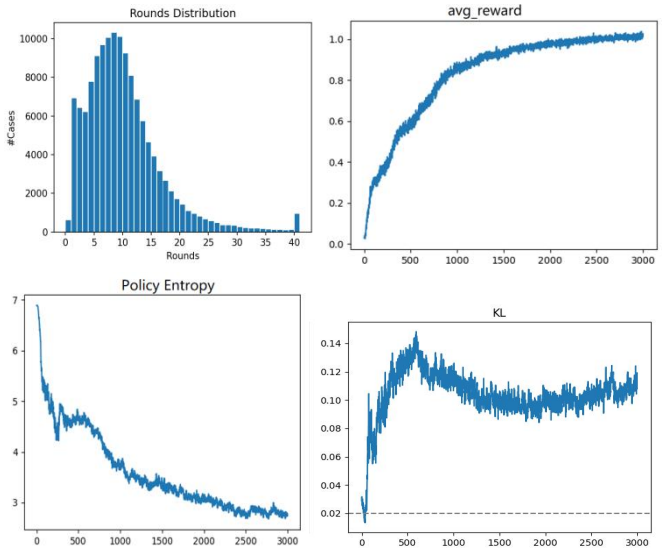


Fig. 16: LSTM performance result

The accuracy rate of the LSTM model astonishingly reached 0.9414, and the F1 score was 0.8660. Average 9.60 round of dialogue round number, is also very efficient.

This model adopts a single LSTM network to handle the dialogue history, and its hidden states are input into Actor, Critic and the auxiliary disease prediction head (AD_Head). The input for each step is the previous evidence index and patient responses. The PPO algorithm (including GAE, clipping target, entropy reward, and dynamic KL divergence control) is adopted for training, and the AD_Head uses the BCE loss.

The effective modeling of dialogue sequences by LSTM, the promoting effect of auxiliary diagnostic tasks in the unified architecture on representation learning, and the support of the PPO algorithm and its advanced training techniques jointly contribute to the outstanding performance of this model. Its extremely high accuracy and dialogue efficiency make it the best-performing model, but its huge RAM consumption is worth noting.Collectively, the organic combination of these core components has allowed this model to achieve exceptionally high accuracy and outstanding dialogue efficiency across various evaluation metrics, positioning it as the current best-performing model.

F. Model Comparative analysis

	XGBoost	Random Forest	MLP	Transformer	LSTM
Accuracy	0.5866	0.4152	0.5975	0.6196	0.9414
Precision	0.6821	0.3051	0.6077	0.6841	0.8609
Recall	0.5544	0.3749	0.5557	0.6155	0.8756
F1-score	0.5435	0.3043	0.5543	0.6165	0.8660
.vg_rounds	34.9993	32.5765	34.9978	8.64	9.6027
vg_reward	0.7395	0.3481	0.6309	0.5816	1.0200
Runtime	451.04min	1d21h1min	183.12min	23h19.15min	1d20h33min
Memory	VRAM= 6554.5 MB RAM= 8410.8 MB	VRAM=55 08.25 MB RAM=224 3.95 MB	VRAM=63 56.5MB RAM=2221 .0MB	VRAM=22154 .5MB RAM=25376.6 MB	VRAM=24164 .5MB RAM=65396. 6 MB

LSTM-PPO model are far ahead, on all diagnostic indicators show remarkable feats of diagnostic ability. This is mainly attributed to the effective utilization of dialogue history by LSTM, the enhancement of representation learning by auxiliary diagnostic tasks, and the stable training of PPO. Transformer - times, PPO model is significantly better than the first three kinds of models based on REINFORCE, embodies the advantages of the Transformer structure and PPO algorithm. On the basis of three REINFORCE model, the MLP and close also outperform XGBoost RF model This shows that to REINFORCE algorithm, stronger predictor based disease (MLP and XGBoost) can lead to better overall performance, and RF poor performance under this setting.

The Transformer-PPO model performs best in terms of dialogue efficiency, with an average of only 8.64 rounds required to complete the diagnosis. This fully demonstrates the efficient information collection ability learned by its strategic network. PPO LSTM model, followed by average 9.60 round, round number is also very efficient. Based on three kinds of model of REINFORCE dialogue round number were significantly higher (about 32-35 wheel), close to the maximum number of wheel to limit, shows that REINFORCE algorithm obviously deficiencies in learning effective dialogue strategy.

The PPO algorithm (Transformer-PPO, PPO-LSTM) generally shows better training dynamics. Its reward curves are usually smoother and the average rewards converged to are also higher. This is attributed to the clipping target, multi-round update and KL divergence control mechanisms of PPO, which effectively enhance the training stability and sample efficiency.

REINFORCE algorithm (XGBoost, RF, MLP model) of the average reward is relatively low, and the curve may be volatile reward, the convergence speed is slow, this strategy is the inherent high variance problem caused by the gradient method.

LSTM effectively processes the dialogue history through its cyclic structure and achieves the best diagnostic effect in combination with auxiliary tasks. The Transformer performs well in context understanding and efficient information integration with its self-attention mechanism, achieving the shortest number of dialogue rounds.

SVD dimension reduction in the application of RF and MLP model, the effect seems to be owing to the different foundation models. For RF, the performance declines significantly; For MLP, the performance is comparable to or slightly better than that of XGBoost without SVD, but the

dialogue efficiency has not been improved. This might be because the dimensionality reduction leads to the loss of some data information, resulting in the model being unable to be trained correctly.

Based on various performance indicators, the LSTM-PPO model achieved an overwhelming advantage in diagnostic accuracy and F1 score, and maintained an extremely high dialogue efficiency (with an average of 9.6 rounds). It was the model with the best comprehensive performance in this experiment. This fully demonstrates the natural advantages of LSTM in serialized dialogue modeling, as well as the effectiveness of the PPO algorithm combined with auxiliary learning tasks in improving the performance of complex systems. It also proves that the mode of integrating ASD and AD and updating parameters round by round will have a better effect than conducting the two parts of ASD and AD separately. Immediately following is the Transformer-PPO model, which stands out the most in terms of dialogue efficiency (with an average of 8.6 rounds), and its diagnostic performance is also significantly better than that of the REINFORCE model, demonstrating the strong ability of the Transformer architecture in context understanding.

However, due to the failure to closely combine the ASD and AD parts, the parameters cannot be updated in real time based on the feedback, resulting in a lower accuracy rate than the LSTM model. However, both of these two types of advanced models are accompanied by relatively high demands for computing resources.

Although the model based on REINFORCE performs poorly in terms of dialogue efficiency, the MLP and XGBoost models among them still have certain competitiveness in diagnostic indicators and relatively low resource consumption, providing an alternative solution for resource-constrained scenarios. The experimental results clearly reveal the trade-off between performance and resource consumption among the selection of model architecture, reinforcement learning algorithms, and the optimization of training techniques.

VI. FUTURE WORK

Although the medical dialogue systems proposed in this study, especially the models based on PO-LSTM and Transformer-PPO, have achieved remarkable results in terms of diagnostic accuracy and dialogue efficiency, there are still several promising research directions and potential improvement Spaces, which are worthy of further exploration in future work:

Current deep learning models (especially Transformer and LSTM) are often regarded as "black boxes" when making decisions. Future research can be dedicated to enhancing the interpretability of the model. For instance, by visualizing through the attention mechanism, generating natural language explanations, or introducing causal inference modules, users (especially clinicians) can understand the reasons why the model makes specific diagnostic or questioning decisions. This is crucial for establishing user trust and promoting the application of the model in clinical practice.

During the dialogue process, the agent can manage and express its uncertainty about the current diagnosis more proactively. Future work can explore the integration of more complex uncertainty quantification methods (such as Bayesian neural networks, Monte Carlo Dropout, etc.) in PolicyNet. Based on this uncertainty assessment, the agent can adopt more targeted strategies. For example, when the uncertainty is high, it can prioritize asking questions that can minimize the uncertainty to the greatest extent, or request assistance from human experts when necessary [20].

The current dialogue strategy may adopt a relatively consistent questioning pattern for all patients. In the future, research can be conducted on how to dynamically adjust dialogue strategies based on patients' specific backgrounds (such as age, gender, past medical history, and comprehension ability) to achieve more personalized and empathetic interactions [18].

Given that outstanding Transformer and LSTM models are often accompanied by high computing resource consumption, exploring model compression techniques (such as knowledge distillation, pruning, and quantization) to develop lightweight versions of dialogue systems is crucial for their deployment in mobile devices or resource-constrained medical institutions. The goal is to significantly reduce the storage and computing requirements of the model while maintaining high performance.

VII. CONCLUSION

This study systematically designed, implemented and evaluated five different hybrid architecture medical dialogue systems, aiming to explore the potential of combining traditional machine learning, deep learning and reinforcement learning methods in enhancing the assistance of automated medical diagnosis. Through a series of experiments, we compared the combination of disease predictors based on XGBoost, random forest, and MLP with the MLP policy network based on REINFORCE algorithm, as well as more advanced end-to-end reinforcement learning models based on Transformer and LSTM.

The experimental results clearly show that the selection of model architecture and reinforcement learning algorithms has a crucial impact on the diagnostic accuracy, dialogue efficiency and training dynamics of the system.

In conclusion, through a systematic comparison of multiple model architectures, this study verified the great potential of deep learning and reinforcement learning, especially the framework that combines sequence modeling and advanced optimization algorithms, in building high-performance medical dialogue systems. The outstanding diagnostic accuracy and high dialogue efficiency demonstrated by the LSTM-PPO model provide a solid foundation and clear direction for the future development of more intelligent and reliable automated medical diagnostic tools. Future work can further explore directions such as knowledge fusion, model interpretability, personalization and lightweight deployment, with the expectation of better applying such advanced technologies to actual clinical scenarios.

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